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Blockchain-integrated AI framework for secure IoT-based digital advertising ecosystems

Yegen Ouyang¹, Enwei Cao^{1*} and Bingfeng Liu¹

*Correspondence:

Enwei Cao
enweicao55@gmail.com; 026535@jcu.edu.cn

¹Department of Management and Economics, Jingdezhen Ceramic university, Jingdezhen, China

Abstract

The rapid growth of IoT-driven digital advertising requires secure, transparent, and intelligent frameworks to resolve issues like data privacy, fraud, and centralized control. This paper introduces a Federated Reinforcement Learning with Blockchain-based Smart Contracts (FRL-BSC) framework that integrates federated reinforcement learning, blockchain, and edge computing to enhance targeted ad delivery. In the proposed approach, IoT devices locally train reinforcement learning agents, while blockchain smart contracts manage access, ensure transparency, and secure immutable records of model updates. Edge nodes aggregate models, reducing latency and improving scalability in large-scale networks. Experimental results demonstrate that FRL-BSC improves security by 93.1%, increases campaign identification accuracy by 8%, lowers the false positive rate (FPR) by 3.8%, enhances local model accuracy by 92.8%, reduces latency by 38%, and improves revenue sharing by 94.9%. These outcomes establish FRL-BSC as a scalable, privacy-preserving, and fraud-resistant solution for next-generation IoT-based digital advertising ecosystems.

Keywords Blockchain, Federated reinforcement learning, Smart contracts, IoT security, Digital advertising, Edge computing

1 Overview

The rise of IoT devices has produced a new reality for digital advertising that is hyper-connected and data-driven [1]. Smart vehicles, wearables, home automation systems, and retail IoT infrastructures all continuously generate enormous amounts of contextual data and behavioural data [2]. These allow brands and marketers to deliver ad content that is highly personalized and dynamic [3]. The rapid expansion of the IoT within digital advertising creates challenges for privacy, security, transparency, and trust [4]. Centralized data collection and handling methods expose user data to unrestricted access, this places the security and privacy of users at risk [5]. The anonymized, opaque interactions between advertisers and publishers create conditions for ad fraud, unfair revenue sharing, and decreased confidence among ALL parties [6]. Further, the vastness and heterogeneous nature of IoT networks require real-time decision-making and low-latency communication for effective use of recognition and other intelligent IoT services



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that typical cloud-based solutions cannot provide [7]. To address some of these issues, this paper proposes an FRL-BSC framework. Within this framework, IoT devices (smart devices) create reinforcement learning agents from user interaction data to optimize ad targeting and placement, without disseminating raw data [8]. Blockchain-based smart contracts track actions that model updates, permissions, or rights, and hold an immutable, verifiable history of transactions, allowing lenders transparency and accountability [9]. Edge computing nodes aggregate the models developed locally, allowing for low-latency scaling of the number of models and enabling skin-in-the-game real-time personalized ad targeting models [10].

Tests show that FRL-BSC works better than conventional centralized techniques because it improves targeting precision, lowers network latency, and builds confidence between advertisers and publishers [11]. Because of this, FRL-BSC provides a scalable, safe, and privacy-preserving framework for next-generation IoT-based advertising ecosystems in areas like smart retail, linked automobiles, and urban IoT infrastructures [12].

1.1 Inspiration

The inspiration for this research emerged from the growing need for smart, secure, and privacy-protecting ad delivery in rapidly expanding IoT networks. Centralized systems already have difficulties with ad fraud, data misuse, and a lack of openness. This project aims to create a dependable, scalable, and effective advertising environment via the amalgamation of federated reinforcement learning and blockchain smart contracts.

1.2 Problem statement

The potential for innovation with IoT, Blockchain, AI, and machine learning is hindered by the current approaches to both digital ecosystems and smart systems, which face problems with security, scalability, and decision-making. Many existing approaches do not attempt to create integrated strategies for anomaly detection, partner identification, or sustainable energy or resource management. This absence of integrated strategies limits the potential of many interconnected digital platforms to provide high-performance, secure and adaptable services for businesses and technological environments that are continually evolving. The manuscript currently introduces several advanced concepts—IoT, AI, Blockchain, Federated Learning (FL), Reinforcement Learning (RL), Smart Contracts, and Edge Computing—which, when presented together, may overwhelm readers. To streamline, the focus should be placed on the core innovation: the FRL-BSC framework, which uniquely combines federated reinforcement learning with blockchain-based smart contracts. Other elements like IoT and edge computing can be positioned as supporting enablers rather than central innovations. By clearly emphasizing that the novelty lies in using blockchain for secure, transparent coordination of federated reinforcement learning in advertising ecosystems, while edge computing and IoT provide scalability and deployment feasibility, the paper will maintain clarity and highlight its main contribution without diluting the narrative.

1.3 Contributions of this paper

- Proposes a novel FRL-BSC framework to ensure secure, transparent, and privacy-preserving ad targeting in large-scale IoT-based digital advertising ecosystems.

- Integrates edge computing and decentralized model aggregation to reduce latency, enhance scalability, and enable real-time personalized ad delivery without exposing sensitive user data to centralized servers.
- Demonstrates significant improvements in ad targeting accuracy, fraud prevention, and trustworthiness through immutable blockchain transactions, transparent revenue sharing, and secure access control mechanisms among advertisers, publishers, and IoT devices.

This paper's remaining sections are organized as follows: The relevant work of the ecosystem for digital advertising is examined in Sect. 2. The approach is described in Sect. 3. Section 4 provides a structure for the assessment. Section 5 discusses and analyses the paper's results. Section 6 concludes up the report with suggestions for further research.

2 Related work

This section compares several different approaches in the digital ecosystems (IoT, Blockchain, and machine learning) to facilitate marketing, security, anomaly detection, and efficiency of operations. This research examines literature over the past year and assesses algorithms, security integration, and performance outcomes. The overview outlines strengths and weaknesses, and potential use cases to assist with the advancement of smart systems, digital business connections, and methods for intelligent decision making.

2.1 Digital service ecosystem

The current research is to comprehend the function and data science applications by SMEs to improve the effectiveness of their internet marketing. This paper employs a comprehensive literature review as its research methodology, utilizing the computer language R for multiple correspondence analysis (MCA) on the data. The analysis of the findings shows that SMEs employ data science in seven cutting-edge ways in their online marketing campaigns, as noted by Saura et al. [13]. Although researchers have begun to investigate the impact of digitalization on many components of business models, they have mostly overlooked its effects on the external connections that organizations cultivate for value generation. This study uses a qualitative analysis of seventeen interviews to develop a two-stage paradigm for the shift to digital business models by Palmié et al. [14].

To develop a digital business strategy, the merchants work with specialized service providers. When firms from the retail and digital-service ecosystems work together to provide a value proposition for end users, a meta-ecosystem develops. The companies (retailers) try to stand out from their rivals in the meta-ecosystem by Gawer et al. [15]. One of the main ways to do this is via face-to-face contacts with the suppliers of products, consumers, and digital service providers. So, digitization doesn't make face-to-face encounters and intimate human relationships unnecessary. This has significant ramifications for scholarly literature and managerial practice by Tekoniemi et al. [16].

2.2 AI with deep learning and blockchain

The flexibility and decision-making skills of the IoT system are improved by integrating deep reinforcement learning (DRL). To improve mobile transmission and ensure secure data exchange in IoT-enabled smart cities, this study looks at the combination

of Blockchain and DRL. IoT networks function more effectively when DRL and Blockchain are integrated because privacy and security are preserved while IoT application systems are grouped and categorized, as noted by Far et al. [17]. The system uses the Interplanetary File System (IPFS) for decentralized storage, and a combination of digital signature algorithm (DSA) and elliptic curve cryptography (ECC) keeps the data safe. A novel equilibrium-enriched Siberian (EES) method is suggested to improve security and make key generation easier. It combines tactics from Siberian tiger optimization (STO) and equilibrium optimization by Kaushik et al. [18].

This paper introduces a unified methodology that employs Deep Neural Networks and Blockchain technology (DNNs-BCT) to improve the identification and prevention of anomalies in IoT settings. The suggested system uses DNNs to intelligently find anomalies by using multi-layer feature extraction and adaptive learning methods. The DNN model learns from IoT traffic statistics to distinguish between normal and abnormal network activity, as described by AR et al. [19]. This helps it find threats like Distributed Denial of Service (DDoS) assaults, malware injections, and insider threats. The IoT is different from typical rule-based Inter Planetary File System (IPFS) since it learns and adapts to new attack patterns all the time, which makes it more accurate and reduces false positives. To ensure data integrity, transparency, and decentralized security, this paper integrates Blockchain technology into the IoT environment by Maurya et al. [20]. Every IoT device keeps track of its behavior on a private blockchain network. This prevents data from being changed, accessed without permission, or having a single point of failure. The Blockchain uses smart contracts to automatically respond to threats, stopping bad behavior right away without the need for human action by Rahman et al. [21].

2.3 Machine learning based digital business ecosystem

In the digital age, businesses from the same or separate market sectors may work together on digital platforms that let them share information. These groups are all part of a Digital Business Ecosystem. However, the proliferation of these groups makes it more challenging to identify economic partners, clients, suppliers, or other groups with similar financial objectives, according to Benramdane et al. [22]. This paper presents a recommendation system deployed on a digital platform, utilizing matching and hybrid supervised machine learning algorithms. This paper provides a comprehensive examination of the system's operation and the challenges faced in data processing that facilitated the identification of commonalities across related businesses. Therefore, want to enhance the comprehension and analysis of the data to enable the appropriate selection of partners by Goel et al. [23].

Enterprises worldwide utilize "machine learning (ML)" to enhance operational efficiency and scalability. A number of "ML" algorithms and AI technologies are increasing popularity in the area of business analytics. As more and more software is used in company operations, the amount of data being created is growing. This is making "ML" more and more critical, as noted by Sanil et al. [24]. Search engines, third-party websites, social media, and mobile apps are just a few of the areas where the enormous field of online advertising has attracted a number of interest. In internet marketing, it's hard to tell how well online campaigns are doing.

Although existing research explores the roles of IoT, Blockchain, deep learning, and machine learning in enhancing marketing, anomaly detection, security, and business

efficiency, a clear research gap remains in integrating these technologies into a unified, adaptive, and scalable digital service ecosystem. Current studies often address isolated use cases such as IoT security with Blockchain, anomaly detection with deep neural networks, or partner selection using ML-based recommendation systems but they lack a holistic framework that can simultaneously manage security, operational efficiency, and dynamic business collaboration in real time. Moreover, while SMEs increasingly adopt data science for marketing and decision-making, the literature provides limited insight into how these advanced technologies can be tailored to their resource constraints and continuously evolving digital business models. This gap highlights the need for research on cross-technology integration, real-time adaptability, and SME-focused scalability to build resilient, intelligent, and sustainable digital ecosystems. This is generally done by looking at user responses via various metrics, like the number of individuals that click on advertisements or register for products, how many people buy things, or how many people provide feedback through online surveys by Gharibshah et al. [25]. A comparison of existing approaches is shown in Table 1.

3 Methodology

The FRL-BSC framework brings together federated reinforcement learning and blockchain technologies to create a digital advertising ecosystem that is IoT-enabled, highly available, transparent, and scalable. It supports privacy-preserving local training, encourages the trustworthy aggregation of updates, and provides immutable model registration. Smart contracts manage incentive allocation, access rights, and dispute resolution. The Blockchain offers real-time, personalized ad delivery, ensuring low latency and high scalability. Edge computing facilitates efficient operation across the diverse environments that comprise the Internet. While technical terms are necessary to convey methodological rigor and experimental details, the insertion of promotional or marketing-style phrases may obscure the scientific contributions and make it challenging for readers to focus on the core research insights. A consistent writing style that prioritizes precise, formal, and objective language would enhance readability, ensure that the technical advancements are clearly communicated, and maintain the credibility of the work within the academic and research community.

3.1 FRL-BSC framework design

The FRL-BSC framework combines FRL with blockchain-enabled smart contracts (BSC) to facilitate a secure, transparent, and privacy-respecting advertising ecosystem for IoT devices. In this framework, IoT devices train local cooperative RL agents. Edge nodes aggregate local models, while the Blockchain records updates, enforces access mechanisms, provides verifiable transactions, and tracks all actions. This enables real-time, personalized ad serving while reducing the threat of fraud and data misuse.

The proposed framework combines IoT, AI, and Blockchain to establish a secure, transparent, and smart digital advertising environment. As IoT devices gather real-time data about user interaction, the publisher at the edge pre-processes the data to address any concerns about what gets passed to the insight hub (e.g., cloud). The Blockchain, with smart contracts, creates immutable records at a minimum to record user consent, ad transactions, and campaign metadata. Cloud-based AI Engines parse consumer

Table 1 The comparison of existing methods

References	Environment	Algorithm/method	Power scheduling	Security	Energy efficiency/effectiveness
[13]	Digital Service Ecosystem	SMEs' use of data science in seven online marketing strategies	X	X	Medium
[14]	Digital Service Ecosystem	Digital business model transform using a two-stage paradigm	X	X	High
[15]	Digital Service Ecosystem	Retail–digital meta-ecosystem collaboration	X	X	Medium
[16]	Digital Service Ecosystem	Human relationship integration in digitization	X	X	Medium
[17]	IoT + AI (Deep RL) + Blockchain	DRL with Blockchain for secure mobile transmission and IoT system categorization	✓	✓	High
[18]	IoT + Blockchain	EES method combining STO and equilibrium optimization for secure key generation	X	✓	Medium
[19]	IoT + AI (DNNs) + Blockchain	DNNs with adaptive learning for anomaly detection in IoT traffic	X	✓	High
[20]	IoT + Blockchain	Blockchain for IoT data integrity, transparency, and decentralized security	X	✓	High
[21]	IoT + Blockchain (Smart Contracts)	Automatic threat response using blockchain smart contracts	✓	✓	Very High
[22]	Digital Business Ecosystem + ML	Hybrid supervised ML algorithms for partner identification on digital platforms	X	X	Medium
[23]	Digital Business Ecosystem + ML	Data analysis for appropriate partner selection	X	X	Medium
[24]	Digital Business Ecosystem + ML	Enterprise efficiency and scalability improvement using ML	X	X	Medium
[25]	Digital Business Ecosystem + ML	ML-based online advertising performance analysis	X	X	High

behaviour to develop and deliver personalized, dynamic advertising (or messages). Federal learning enhances accuracy without needing to centralize sensitive data sets [26]. Ad exchanges can conduct safe, fraud-resistant bidding practices while becoming the payment and audit module to add transparency to ad deployment. Implementing all these technologies can drive trust, efficiency, and privacy through new advertising technologies that form new-generation advertising networks, as illustrated in Fig. 1.

Joint utility with on-chain verification & privacy leakage penalty $\epsilon(y, t)$ is expressed using Eq. 1,

$$\epsilon(y, t) = \partial_v(y_v, t_v) * (-\Delta H(t)) \quad (1)$$

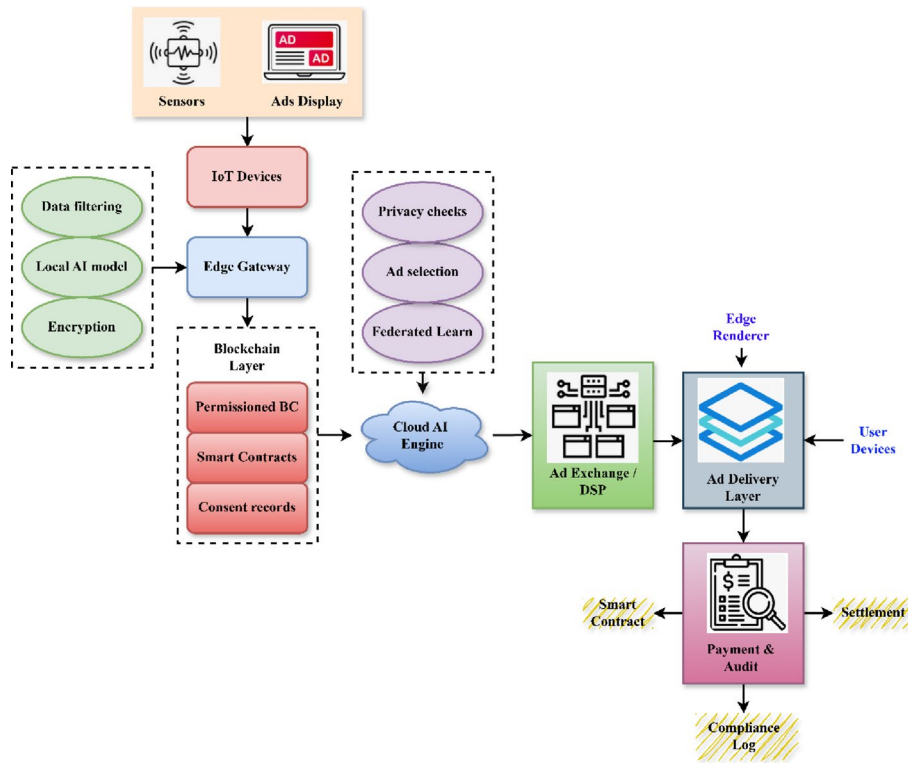


Fig. 1 Blockchain-integrated AI framework for secure IoT-based digital advertising ecosystems

Equation 1 explains the joint utility with on-chain verification & privacy leakage penalty, high chain saturation nonlinearly diminishes the effective impact [27]. In this $\epsilon(y, t)$ is the scalar joint objective, y is the allocation intensity to IoT endpoints, t is the signed verification tuples, $\partial_v(y_v, t_v)$ is the idiosyncratic impact kernel for the endpoint, $(-\Delta H(t))$ is the attenuation factor, and $H(t)$ is the aggregate on-chain resource function.

Byzantine-resilient reputation update with stake-weighted verification and slashing S_j^{u+1} is expressed using Eq. 2,

$$S_j^{u+1} = \mathbb{C}(S_j^u + \beta * \theta_j^u - \delta) \quad (2)$$

Equation 2 explains the Byzantine-resilient reputation update with stake-weighted verification and slashing via a sigmoidal gain from the local verification score.

In this S_j^u, S_j^{u+1} are the scalar reputation of node at times, $\mathbb{C}$ is the squashing map ensuring reputations remain bounded, β is the learning/gain coefficient for positive reinforcement, θ_j^u is the veracity score for node at time, δ is the slashing intensity scalar converting on-chain penalties into reputation decrease, j is the node index, and u is the discrete epoch index.

Input:

S_j^u // Current reputation of node j at time u

θ_j^u // Veracity score of node j at time u

β // Learning/gain coefficient (positive reinforcement)

δ // Slashing intensity scalar (penalty factor)

C // Squashing function (to keep reputation bounded)

Output:

S_j^{u+1} // Updated reputation of node j at time $u + 1$

Procedure UpdateReputation($S_j^u, \theta_j^u, \beta, \delta, C$):

Step 1: Calculate raw reputation update value

$$raw_{update} = S_j^u + \beta * \theta_j^u -$$

Step 2: Apply squashing function C to ensure bounded reputation

$$S_j^{u+1} = Squash(raw_{update})$$

return S_j^{u+1}

Function Squash(value):

Sigmoidal squashing map example to bound reputation between 0 and 1

Can be replaced by any appropriate bounded mapping function

if $value < 0$:

return 0

else if $value > 1$:

return 1

else:

return value

Algorithm 1 Evaluation of Byzantine-Resilient Reputation

The Algorithm 1, a node's reputation by combining its current score, positive reinforcement from verification, and penalties from slashing. It first calculates a raw update by adding the product of the learning coefficient (β) and the node's veracity score (θ), then subtracts the slashing penalty (δ). To ensure reputations remain within valid bounds, a squashing function is applied typically a sigmoidal or bounded map that limits the reputation to a predefined range (e.g., 0 to 1). This process makes the reputation

system resilient to Byzantine faults by rewarding honest behavior and penalizing malicious actions while keeping scores stable and bounded.

The complete FRL-BSC ecosystem, highlighting the interaction (the flow) from the IoT devices into Blockchain and Cloud Analytics. The IoT clusters generate telemetry data and local reinforcement learning decisions, then pass updates to the Edge Orchestrators. There, aggregating occurs, and temporary storage takes place while the in-cloud version waits to complete the transaction. Meanwhile, the smart contracts in Blockchain take care of suspicious model development, revenue sharing, dispute resolution, and are decentralized for model, technical security, and transactional transparency. An earlier step of off-chain storage (in an immutable format) is done for larger model files. At the same time, verification occurs for the models using the on-chain record of the metadata attribute, model attributes, or hashes of those attributes. Cloud will provide the long-range retraining, model assessments, and advertising strategic analysis [18]. Adopter/Publisher interfaces will offer in-context dashboards for their campaign performance and payment. This ecosystem landscape will need to find the equilibrium of decentralization, performance, privacy, and trust above all else in Fig. 2.

Data throughput balance across pipeline stages α_{ot} is expressed using Eq. 3,

$$\alpha_{ot} = \alpha_{sr}, \frac{\alpha_{pr}}{\sigma_{lt}}, \alpha_{cn} \quad (3)$$

Equation 3 explains the data throughput balance across pipeline stages slowest section of source reflection, latency-adjusted interpreting, and channel transmission limits.

In this α_{ot} is the steady-state output data rate, α_{sr} is the raw emission rate from the source subsystem, α_{pr} is the nominal processing rate in the computational stage, σ_{lt} is the latency amplification factor, and α_{cn} is the maximum channel transmission rate.

End-to-end delay accumulation δ_{ot} is expressed using Eq. 4,

$$\delta_{ot} = \delta_{sr} + \delta_{pc} + \delta_{tn} + \delta_{snc} \quad (4)$$

Equation 4 explains that the end-to-end delay accumulation source capture delay, calculation delay, transfer delay, and sync delay represent the total operation or frame delay.

In this δ_{ot} is the total end-to-end latency, δ_{sr} is the acquisition/capture delay at the source, δ_{pc} is the computation processing delay, δ_{tn} is the transmission delay, and δ_{snc} is the synchronization reassembly delay.

Energy cost of end-to-end operation F_{ot} is expressed using Eq. 5,

$$F_{ot} = F_{sr} + \tau_{pr} * F_{pr} + \tau_{cn} * F_{tn} \quad (5)$$

Equation 5 First, the source sector energy accounts for the baseline energy consumed by the devices or nodes generating data, which includes powering sensors, computation units, and storage. Next, the process energy represents the additional energy required to process or compute the data, which is influenced by the efficiency of processing factor—a measure of how effectively the system converts electrical power into computational output. Higher processing efficiency reduces the energy overhead per operation, whereas lower efficiency increases it proportionally. Finally, the transmitting energy quantifies the energy consumed in transmitting data across the network, and this is typically scaled by network parameters such as the number of hops, transmission distance, data size, and communication protocol overhead.

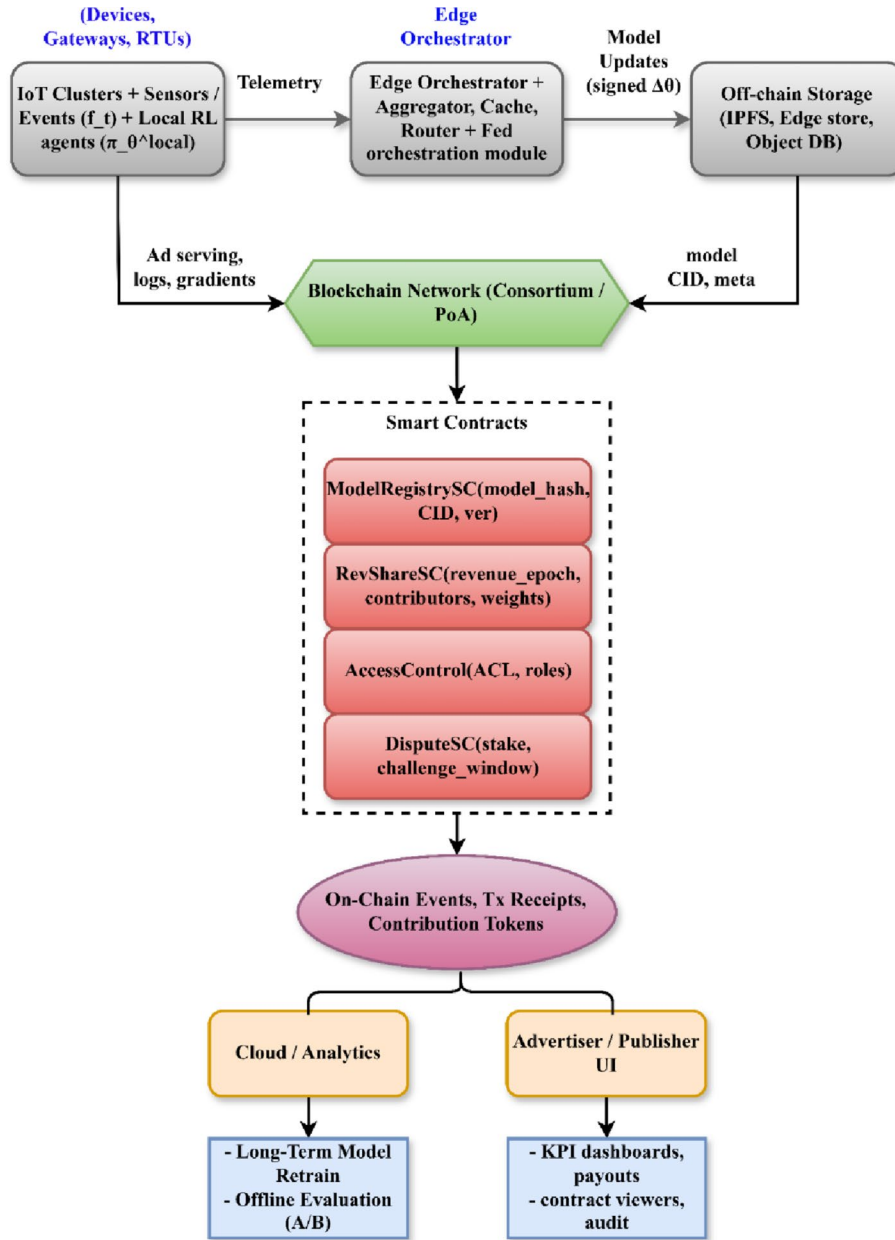


Fig. 2 System-level end-to-end architecture

In this F_{ot} is the total energy consumption for the end-to-end transaction, F_{sr} is the energy consumed by the source subsystem, F_{pr} is the energy consumed during processing, F_{tn} is the energy consumed during transmission, τ_{pr} is the processing efficiency factor, and τ_{cn} is the channel efficiency factor [20].

3.2 Trust & transparency mechanisms

The FRL-BSC framework provides assurance required for trust and transparency in IoT advertising based on a blockchain's immutable ledger, which retains all model updates, transactions, and revenue-sharing agreements. Smart contracts assist with executing compliance requirements and enforce specifications, requirements, and permissions with no need for a mediator. Cryptographic methods verify and protect these systems

against tampering. Decentralized consensus obviates the weaknesses that centralization can create, enabling a more trustworthy, auditable, and fraud-free IoT advertising ecosystem.

In the context of the FRL-BSC framework, each IoT device follows a workflow. The devices gather contextual information, produce feature vectors, and implement reinforcement learning policies that allow the devices to choose actions (e.g., the delivery of a targeted ad). Next, the devices compute rewards and retain the state–action–reward sequences within the local replay buffer, using the weighted engagement metrics from the required data and relevant user experience measures. The device creates model updates at predetermined intervals, following gradient compression and (optionally) the addition of a differential privacy noise. The devices digitally sign the updates with their keys, which are then sent securely to the edge orchestrator. This localized loop allows for learning confined to the specific devices, serving a role of privacy-preserving improvements while adding to the improvements of the federated model process in Fig. 3. The manuscript presents a Byzantine-resilient reputation algorithm, outlining its design and intended ability to tolerate malicious or faulty nodes within the network. However, the framework is not rigorously validated through systematic attack simulations, such as Sybil, collusion, or spoofing attacks, which are critical for assessing its true resilience. Without such empirical validation, it remains unclear how effectively the algorithm can

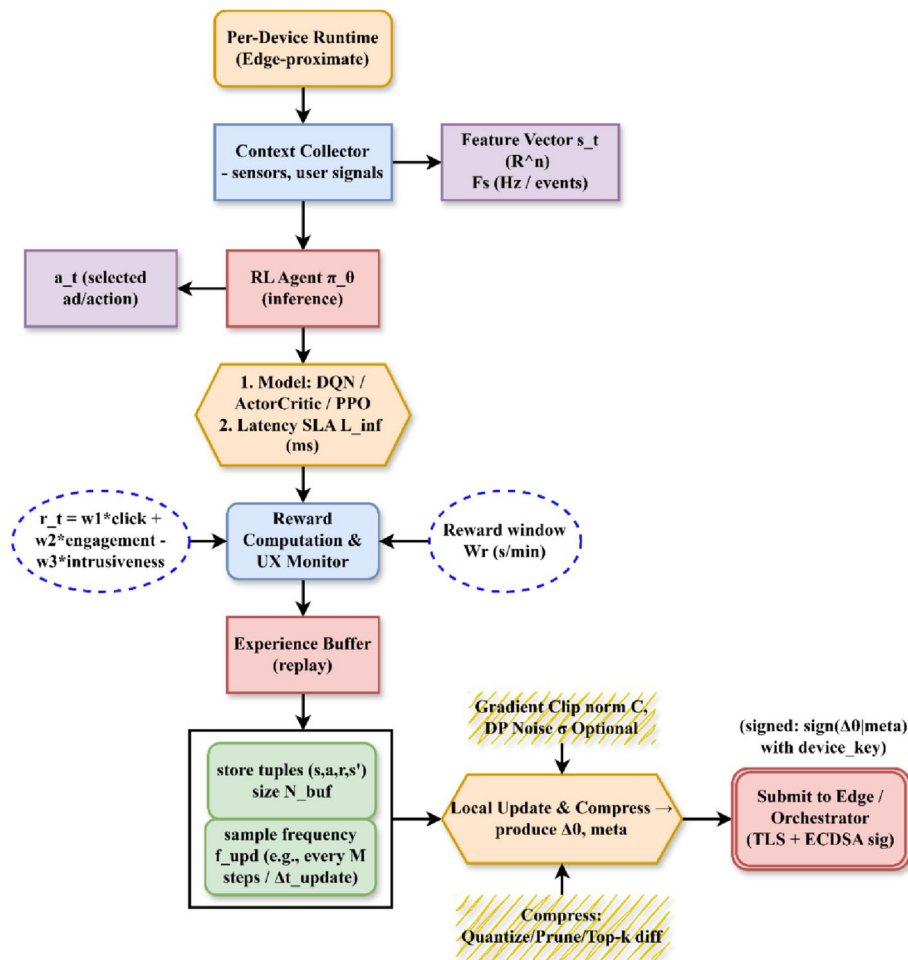


Fig. 3 Per-device flow of RL agent training, serving, and submission

detect and mitigate malicious behavior, or how its performance scales under varying levels of adversarial participation. Incorporating comprehensive simulation studies would strengthen the credibility of the proposed approach and provide quantitative evidence of its robustness in realistic, adversarial network conditions.

Local policy parameter update $x_e^{(u+1)}$ is expressed using Eq. 6,

$$x_e^{(u+1)} = x_e^{(u)} + \partial_e \Delta_x K_e \quad (6)$$

Equation 6 explains the local policy parameter update using the data batch, the device d changes its policy parameters over its local performance function.

In this $x_e^{(u)}$ is the policy parameter vector for device, $x_e^{(u+1)}$ is the updated policy parameter vector at step, ∂_e is the device-specific learning coefficient, Δ_x is the gradient operator with respect, K_e is the scalar performance objective for device, e is the device index, and u is the local training iteration index.

Aggregated submission to the global model $X_{gbl}^{(U)}$ is expressed using Eq. 7,

$$X_{gbl}^{(U)} = \sigma_e x_e^{(U)} \quad (7)$$

Equation 7 explains the aggregated submission to the global model each device submits its parameters, weighted by its dependability factor, into a balanced global aggregate at submission time.

In this $X_{gbl}^{(U)}$ is the aggregated global parameter vector at the submission round, $x_e^{(U)}$ is the final parameter vector from device, σ_e is the device reliability weight, and U is the final training submission round index.

The edge nodes are responsible for managing the federated reinforcement learning rounds within FRL-BSC. The federated learning ensures user privacy, it provides limited detail on the specific mechanisms employed to achieve this protection. Important strategies including encryption methods, safe multi-party computing, and differential privacy, or data anonymization strategies are not thoroughly discussed, leaving uncertainty about the extent to which sensitive information is safeguarded. Without explicit implementation details or evaluation of these privacy-preserving methods, it is difficult to assess the framework's effectiveness against potential data leakage, inference attacks, or reconstruction threats. A deeper exploration of these privacy mechanisms, along with experimental validation, would significantly strengthen the credibility of the privacy claims and demonstrate compliance with rigorous data protection standards. The rounds initiate with a round initialization, where the devices provide encrypted updates of their models to the edge nodes. The edge nodes then verify, however they can, if the updates are valid, given the cryptographic signatures, and any anomalously generated updates are filtered. Each device's contributions have their quality and performance impact, given a score. Then those contributions are aggregated using techniques such as FedAvg or FedProx, in cases where there are weighted contributions of more or less than n (n is set in the design). A performance comparison that meets performance and fairness thresholds will validate the updated global model with datasets set aside for the edge-side evaluation. The edge side will set aside any models deemed successful off-chain and register successful models on-chain. The edge nodes will broadcast the successful models to the registered devices. If quality metrics fall outside thresholds, rollback protocols are adopted, providing stability for quality and trustworthy deployments in Fig. 4.

Local edge update $Q_f^{(l+1)}$ is expressed using Eq. 8,

$$Q_f^{(l+1)} = Q_f^{(l)} + \partial_f \Delta_q S_f \quad (8)$$

Equation 8 explains the local edge update every edge node, modifies its local policy vector by moving up the incline of its reward functional, over the most recent interactions.

In this $Q_f^{(l)}$ is the policy parameter vector at iteration, $Q_f^{(l+1)}$ is the updated parameter vector at iteration, ∂_f is the learning coefficient for edge, Δ_q is the gradient operator, S_f is the local reward-based objective functional for edge, f is the edge node index, and l is the local training iteration counter.

Edge-to-aggregator parameter transmission $Q_{ag}^{(s)}$ is expressed using Eq. 9,

$$Q_{ag}^{(s)} = x_f Q_f^{(s)} \quad (9)$$

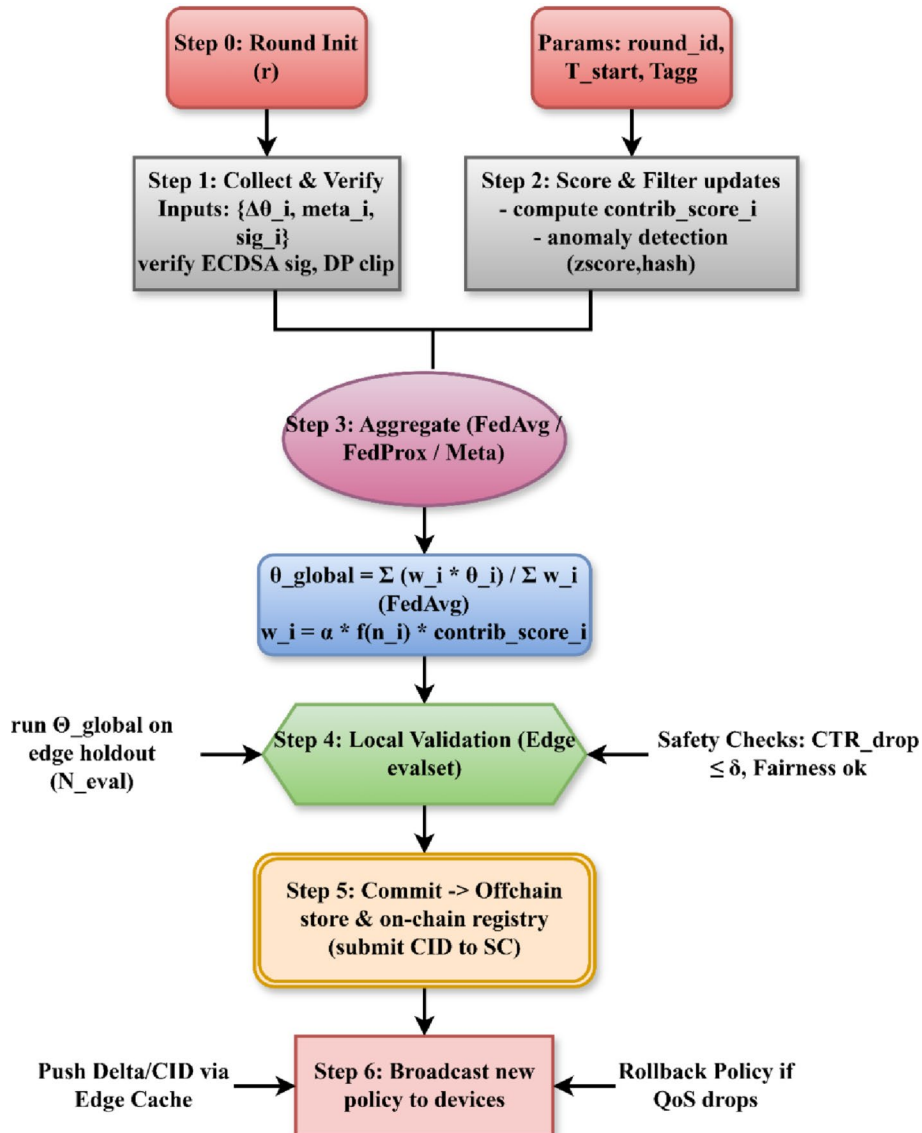


Fig. 4 Edge aggregation & federated RL cycle

Equation 9 explains the edge-to-aggregator parameter transmission to calculate a consensus parameter set, the aggregation tool receives parameter vectors.

In this $Q_{ag}^{(s)}$ is the aggregated parameter vector at round, $Q_f^{(s)}$ is the parameter vector from edge in round, x_f is the reliability or importance weight for edge, and s is the federated aggregation round index.

Global reward evaluation $\bar{S}^{(s)}$ is expressed using Eq. 10,

$$\bar{S}^{(s)} = \frac{1}{|\circ F|} S_f \left(Q_{ag}^{(s)}, Y_f^{(s)} \right) \quad (10)$$

Equation 10 explains the global reward evaluation as the aggregated parameters are assessed over all edges' datasets, the total reward metric is calculated as a median of local rewards.

In this $\bar{S}^{(s)}$ is the mean reward across all edges for round, $|\circ F|$ is the total number of edge nodes in the federation, S_f is the reward functional for edge, $Q_{ag}^{(s)}$ is the aggregated parameter vector for round, $Y_f^{(s)}$ is the evaluation batch for edge at round, $\circ F$ is the set of all edge nodes, and s is the federated aggregation round index.

3.3 Edge-enhanced scalability

The FRL-BSC framework employs edge-compute aggregation of local reinforcement learning models, minimizing transported data and delay latency. Processing data nearer to an IoT device and then optimizing Ad-delivery enhances real-time personalization at scale. In summary, having greater headroom with local ad delivery enables more scaling, network and environment efficiency, supports privacy, and reduces workload on a central server.

The blockchain smart contracts in FRL-BSC. The model registry contract caches new model updates and includes both a cryptographic hash of the model update and an off-chain content identifier to enable provenance and immutability. Contributors are allocated tokens in proportion to the resulting contributions related to their model, and this is managed through the revenue share contract. Access control contracts specify and regulate the retrieval rights for encrypted model files, and the dispute contract allows challenges to submissions deemed to be suspicious; this will be supported by a staking and slashing mechanism to deter fraud. Off-chain storage systems like IPFS will handle relatively larger encrypted model binaries, and only proofs and metadata will remain on the chain. This provides guaranteed transparent incentives, secured access, and credible fraud deterrents, as illustrated in Fig. 5. IoT-based advertising systems demand extremely low-latency responses, often at the millisecond level, to ensure timely ad delivery and maintain user engagement. While the framework proposes mechanisms for data processing and decision-making at the edge, its real-time feasibility has not been thoroughly evaluated. Specifically, the manuscript does not provide detailed measurements of end-to-end latency, including computation, communication, and model inference delays under varying network loads and device densities. Without such analysis, it is unclear whether the system can consistently meet the stringent responsiveness requirements of high-speed IoT environments, limiting confidence in its practical deployment for real-time advertising applications.

Revenue distribution per model instance $\Delta_n^{(u)}$ is expressed using Eq. 11,

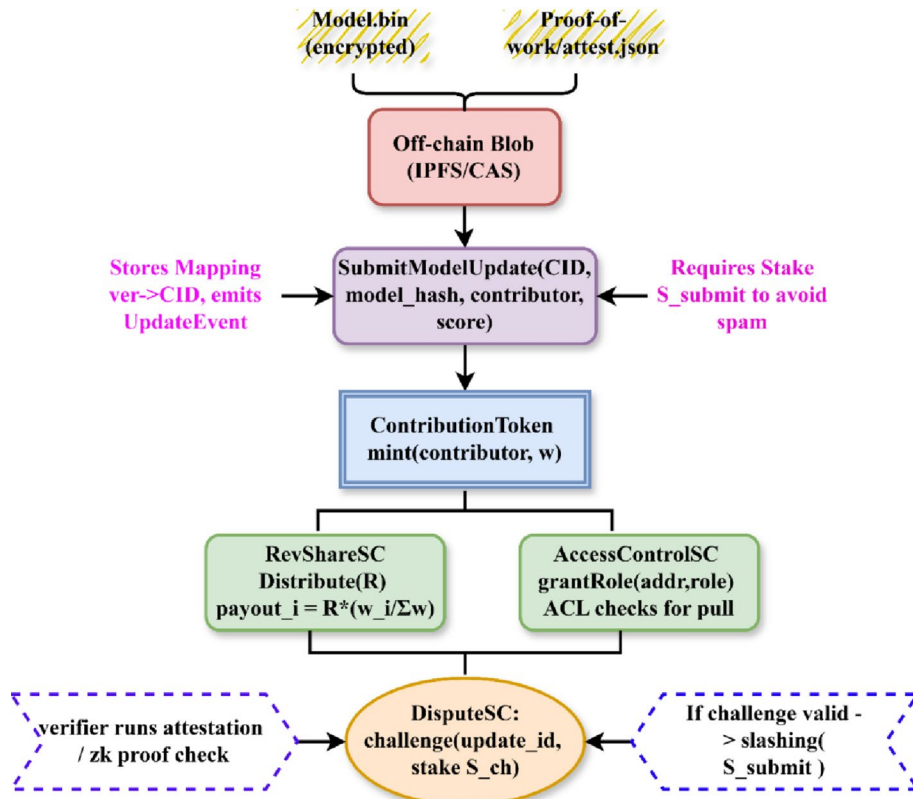


Fig. 5 Smart contract lifecycle for model registry, access, revenue & dispute flow

$$\Delta_n^{(u)} = \rho_n * \sigma_n^{(u)} * G^{(u)} \quad (11)$$

Equation 11 explains the revenue distribution per model instance dividend for a model at a lifecycle epoch is equal to its stake times its confirmed use rate.

In this $\Delta_n^{(u)}$ is the revenue assigned to the model at epoch, ρ_n is the locked stake or collateral for model, $\sigma_n^{(u)}$ is the verified access usage rate for the model during epoch, $G^{(u)}$ is the total revenue pool for distribution at epoch, u is the discrete smart contract lifecycle epoch, and n is the model index in registry.

Dispute resolution outcome $\nabla_n^{(u)}$ is expressed using Eq. 12,

$$\nabla_n^{(u)} = l_{vr} * \mathbb{I}(\tau_n^{(u)}) * \rho_n \quad (12)$$

Equation 12 explains the dispute resolution outcome a punishment that is generated by disagreement if the model's verification score.

In this $\nabla_n^{(u)}$ is the stake penalty imposed on model at epoch, l_{vr} is the penalty coefficient for failed verification, $\mathbb{I}(\cdot)$ is the indicator function, $\tau_n^{(u)}$ is the verification score for model, ρ_n is the locked stake for model, u is the discrete smart contract lifecycle epoch, and n is the model index in registry.

In summary, the FRL-BSC technology improves IoT advertising networks by combining AI-driven customization with blockchain-based trust. Local device learning keeps the information private, while edge aggregation speeds up performance. Blockchain smart contracts ensure that model monitoring is transparent, income sharing is

equitable, and fraud is prevented. This design makes ad delivery scalable, fast, and safe, which helps the whole ecosystem be more reliable, efficient, and accountable.

4 Evaluation setup

4.1 Dataset description

The Digital marketing Ads & Events sample dataset provides a comprehensive overview of how well eCommerce ads are performing. It records visitors, session data, ad interactions, and campaign information from several ad campaigns, which add up to about 10 billion events per year. Great for understanding how people interact with ads, improving campaign ideas, and aiding in machine learning jobs within marketing analytics and customization frameworks [28]. Table 2 presents the simulation environment.

4.2 Simulation environment

See Table 2.

4.3 Evaluation metrics

Evaluating the FRL-BSC framework in a thorough and meaningful manner necessitates the use of well-defined and precise performance metrics that capture multiple dimensions of effectiveness. These metrics are essential not only for assessing technical performance but also for understanding the practical impact of the framework on all stakeholders. In particular, security metrics gauge the framework’s resilience against attacks, unauthorized access, and potential manipulation, ensuring the integrity of the advertising ecosystem. Accuracy metrics measure the ability of the system to correctly identify and classify ad campaigns, while the false positive rate (FPR) provides insight into the frequency of incorrect identifications, which can directly affect decision-making and revenue allocation. Additionally, local model accuracy reflects the reliability of individual participants’ contributions in the federated setting, and efficiency metrics such as latency evaluate the computational and communication overhead, which is crucial for real-time ad delivery. Finally, fairness metrics, including the fair distribution of revenue, assess whether all participants, including advertisers, publishers, and platform providers,

Table 2 The simulation environment

Component	Description	Purpose
IoT device agents	Simulated IoT devices (e.g., connected vehicles, smart retail sensors) that locally collect user data and train reinforcement learning agents	Enables decentralized, privacy-preserving local model training and personalized ad targeting
Blockchain network	Distributed ledger maintains immutable records of model updates, transactions, and smart contract executions	Ensures transparency, security, and trust among advertisers, publishers, and IoT devices
Smart contracts	Automated protocols that enforce access control, revenue sharing, and validate transactions	Guarantees fair and verifiable interactions, prevents fraud, and automates payments
Ad inventory simulator	Virtual pool of ads with metadata (target demographics, budgets, formats)	Provides realistic ad content and campaign options for placement and targeting decisions
User interaction model	Simulated user behavior patterns, preferences, and responses to ads on IoT devices	Tests the effectiveness, engagement, and feedback for reinforcement learning optimization
Network environment	Simulates communication delays, bandwidth limitations, and potential attack scenarios in IoT networks	Evaluates latency, security resilience, and system robustness under realistic network conditions

benefit proportionally from the system's operations. By combining these dimensions, the evaluation framework offers a comprehensive, holistic view of FRL-BSC's effectiveness, ensuring that improvements in security, accuracy, efficiency, and trust are all systematically quantified and balanced.

Analysis of security mechanism systemic robustness index δ_{sc} is expressed using Eq. 13,

$$\delta_{sc} = (\partial_d * l_d) * f^{-\varnothing * w} \quad (13)$$

Equation 13 explains the analysis of the security mechanism systemic robustness index by aggregating per-component hardening terms, the index attenuates vulnerability load.

In this δ_{sc} is the systemic robustness index, ∂_d is the intrinsic tamper-resistance factor of component, l_d is the correctness/coverage coefficient for component, w is the aggregate vulnerability load, $\varnothing$ is the vulnerability sensitivity scalar, and $f^{-\varnothing * w}$ is the exponential decay factor reflecting compounded risk.

Analysis of ad campaign id identifier distinctiveness quotient ∇_{ad} is expressed using Eq. 14,

$$\nabla_{ad} = \frac{\Sigma_{rch}}{1 + y_{cl}} \quad (14)$$

Equation 14 explains the analysis of ad campaign id identifier distinctiveness quotient penalizes re-used or ambiguous ids and adjusts the advertising effort's reach.

In this ∇_{ad} is the identifier distinctiveness quotient, σ_{rch} is the measured reach attributable to that campaign id, and y_{cl} is the collision metric.

Analysis of false positive rate weighted FPR ∂_{FQS} is expressed using Eq. 15,

$$\partial_{FQS} = X_j * 1 \{ \hat{z}_j = 1, z_j = 0 \} \quad (15)$$

Equation 15 explains the analysis of false positive rate weighted FPR when counting false positives, weighted employs sample weights to account for significance.

In this ∂_{FQS} is the weighted false positive rate, j is the sample or event index across evaluation set, x_j is the nonnegative importance weight for sample, $\hat{z}_j$ is the predicted positive indicator, z_j is the ground-truth label, and $1\{\cdot\}$ is the indicator function.

Analysis of local model update local drift magnitude ∇_e is expressed using Eq. 16,

$$\nabla_e = {}^\circ F_e \|V_e\|_2 \quad (16)$$

Equation 16 explains the analysis of local model update local drift magnitude scaled by device-specific adaptation quantifies the size of the device's update vector.

In this ∇_e is the scalar drift magnitude for device, ${}^\circ F_e$ is the device adaptivity/scaling factor, v_e is the raw parameter update vector produced locally on device, and $\|\cdot\|_2$ is the euclidean norm.

The analysis of network latency effective transport delay a_{nt} is expressed using Eq. 17,

$$A_{nt} = \Sigma_{rr} + \frac{\Phi_r}{1 + \delta_{cx}} \quad (17)$$

Equation 17 explains the analysis of network latency effective transport delay latency in effect baseline, and a queue-term, attenuated by the available bandwidth factor are added together to form.

In this α_{nt} is the effective network latency, σ_{rr} is the baseline round-trip time, ϕ_r is the queued delay estimate, and δ_{cx} is the normalized available bandwidth factor.

The analysis of revenue share stake & usage weighted split β_t is expressed using Eq. 18,

$$B_t = P_t * (\Sigma_t)^Y * S \quad (18)$$

Equation 18 explains the analysis of revenue share stake & usage weighted split a portion of revenue is allocated the pool proportionately to stake and consumption.

In this β_t is the revenue allocated to provider, S is the total distributable revenue pool, ρ_t is the locked stake or weight, σ_t is the observed verified usage metric, and Y is the usage sensitivity exponent.

Blockchain transaction hash uniqueness confidence β_{uy} is expressed using Eq. 19,

$$B_{uy} = 1 - \frac{o_{uy}(O_{uy} - 1)}{2 * 2^c} \quad (19)$$

Equation 19 explains the blockchain transaction hash uniqueness confidence good when the estimated collision chance from the natal bound is the subtraction term.

In this β_{uy} is the uniqueness confidence, o_{uy} is the number of transactions under consideration in the pool, and c is the hash output length in bits.

User interaction data composite engagement signal e_{en} is expressed using Eq. 20,

$$E_{en} = \frac{\partial_{cl} * D + \partial_{\mathcal{C}} * U + \partial_t * T}{F + \vartheta} \quad (20)$$

Equation 20 explains that the user interaction data composite engagement signal per-exposure engagement is calculated by normalizing composite engagement by exposures.

In this e_{en} is the per-exposure engagement score, D is the count of clicks, U is the aggregate dwell time or normalized time metric, T is the aggregate scroll interaction depth metric, ∂_{cl} , $\partial_{\mathcal{C}}$, ∂_t are the importance weights for click, time, and scroll respectively, F is the number of exposures/impressions, and ϑ is the tiny positive constant to avoid division by zero.

The evaluation metrics show that substantial improvements have been made in all the proposed measures of performance including: 93.1% in security; 8% improvement with it comes to the accurate identification of ad campaigns; 3.8% better FPR; local model accuracy achieved 92.8%; the network latency of the solution was reduced by 38%; and it also had a gain of 94.9% in the rate of revenue sharing. Overall evaluation metrics provide evidence that FRL-BSC provides high levels of security, efficiency, and trust for IoT-based digital advertising.

5 Results and discussion

This paper assesses the Blockchain-Integrated AI Framework for Secure IoT-Based Digital Advertising Ecosystems' performance across multiple dimensions. The current study evaluated the performance of the proposed FRL-BSC, MCA, DNNs-BCT, and IPFS based on the measures of security, efficiency, and accuracy. In addition to comparing the performance of the security mechanisms, the evaluation covered the monitoring of

the ad campaign to facilitate user interactions, false positives, model updates, blockchain transactions, network latency, and payment records to assess performance for a more holistic performance of the entire system.

Security mechanisms employed in IoT-based advertising ecosystems are responsible for protection against unauthorized access, data breaches, and even fraud. Security mechanisms employ methods like anomaly detection in advertising campaigns, encrypt all data transferred during advertising transactions, and utilize authentication protocols to safeguard user data and transactions related to advertisements. While the immutable ledger ensures the security of historical advertising interactions through the Blockchain, federated learning secures model updates via secure connections among devices. It leverages each user's context to 93.1% made evaluated using Eq. 13. IPFS maintains relatively high but fluctuating performance, suggesting reliable data integrity yet inconsistent detection strength. In contrast, FRL-BSC consistently outperforms the others, showing steady improvement and achieving the highest security event count by interval 8, highlighting its robustness and adaptability in evolving digital ecosystems. Overall, FRL-BSC emerges as the most effective and sustainable approach for security event management. The resiliency of both methods combined further incentivizes trust among users and advertising agencies, which is critical for the success of large-scale digital advertising endeavors that prioritize user privacy and transparency in advertising transactions, as illustrated in Fig. 6.

Ad Campaign ID denotes the advantage of identifying specific advertising campaigns within the IoT ecosystem. Having an Ad Campaign ID enables the tracking and reporting of multiple advertising campaigns concurrently, targeting different user groups

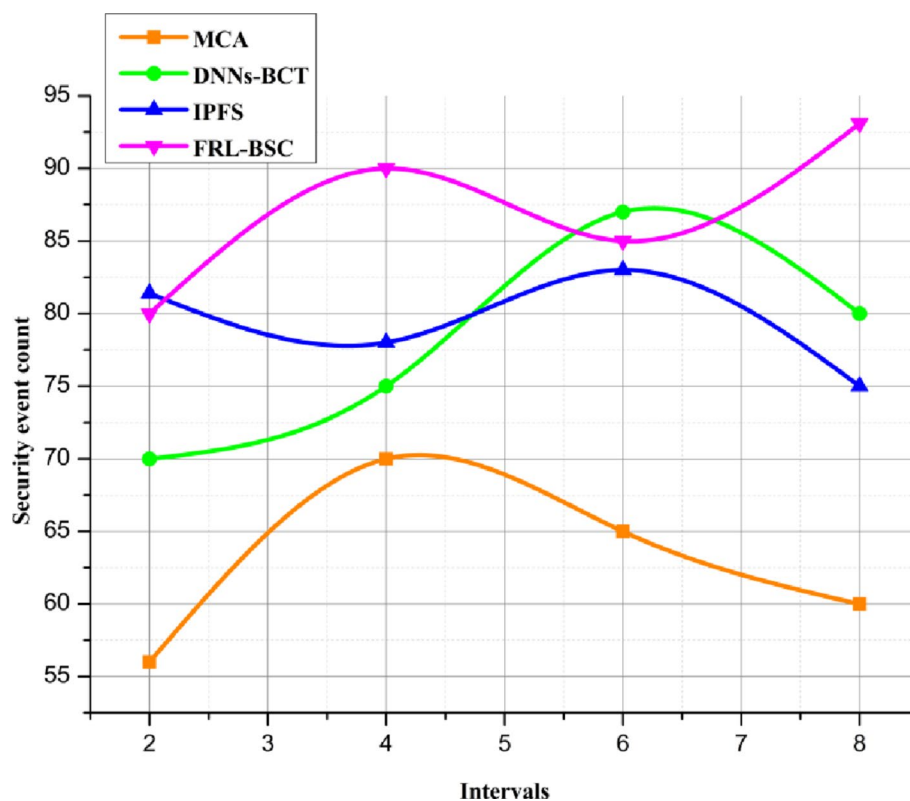


Fig. 6 The analysis of security mechanism

across various devices. This provides insight into user groups, optimizes ad content for users' personal context, and addresses the dependencies of advertisers and publishers. In FRL-BSC, the Ad Campaign ID serves as a value that links Blockchain and FRL recorded transactions and algorithmic decisions to specific campaigns made evaluated using Eq. 14. The bar chart compares the number of impressions across four devices for MCA, DNNs-BCT, IPFS, and FRL-BSC. For Device_001, DNNs-BCT achieves the highest (6), followed by IPFS (5), MCA (4), and FRL-BSC (4). In Device_002, FRL-BSC leads with 7 impressions, IPFS records 6, DNNs-BCT 5, and MCA 4. For Device_003, DNNs-BCT peaks at 7, FRL-BSC follows with 6, while both IPFS and MCA register 5. Finally, in Device_004, FRL-BSC dominates with 8, DNNs-BCT records 5, MCA 4, and IPFS the lowest at 3. Overall, FRL-BSC consistently shows higher performance across devices. This facilitates reporting and auditing on a broad scale, as well as ensuring an equitable revenue share among advertisers and publishers, at a rate of 8%. The parameter of Ad Campaign ID is vital in a system that delivers advertisements in a targeted way, optimizes personalized content effectively, and most importantly, maintains trust between advertisers and publishers within the federated learning of evolving and decentralized users in Fig. 7.

The user interaction data in FRL-BSC enables continued optimizations of IoT advertising. It consists of an interaction timestamp (synchronized with the training and blockchain logging), Device ID (to enable anonymous device-level tracking), Ad ID (for attribution of performance), and interaction type (to inform what rewards to generate via reinforcement learning); this information ensures accurate targeting, adheres to privacy preservation, and allows for transparency of revenue sharing in decentralized advertising ecosystems in Table 3 made evaluated using Eq. 15.

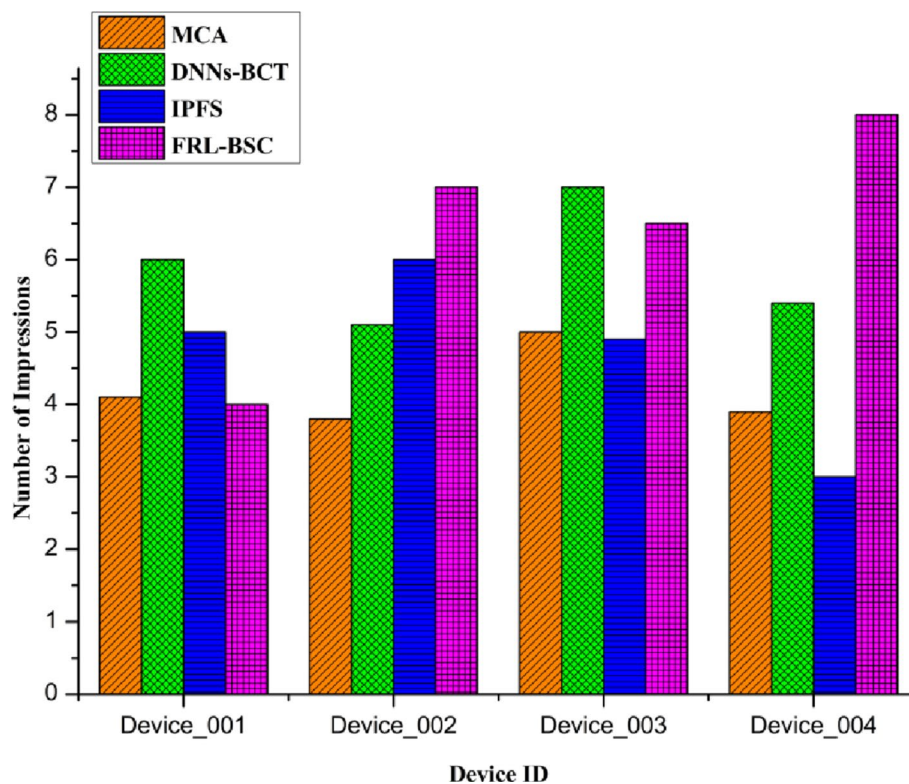
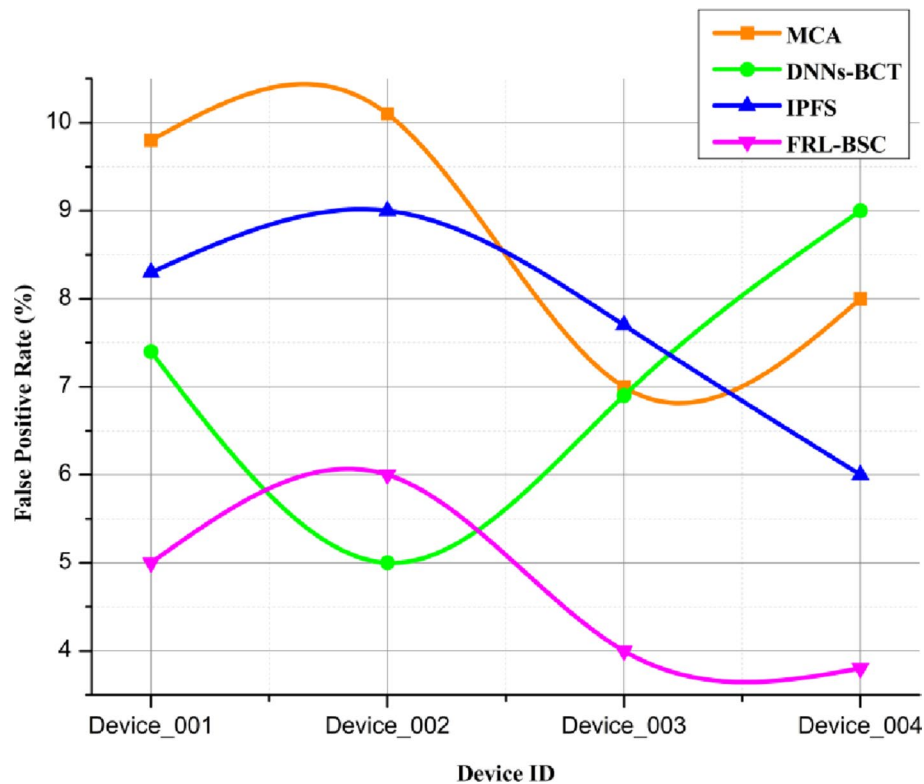


Fig. 7 The analysis of Ad campaign ID

Table 3 User interaction data

Attribute	Description	Data type	Example value	Usage in FRL-BSC
Interaction Timestamp	Time of user engagement with the ad	DateTime	2025-08-11 14:22:35	Synchronizes local RL training and blockchain transaction recording
Device ID	Unique identifier for the IoT device	String	IoT-Retail-3421	Links interaction data to device without revealing personal identity
Ad ID	Identifier of the ad shown to the user	String	AD_90832_XYZ	Used in smart contracts for revenue attribution and ad performance measurement.
Interaction Type	Nature of user engagement with the ad	Enum	Click, View	Determines RL reward value and informs personalized ad targeting

**Fig. 8** The analysis of false positive rate

For a low FPR to happen, it's important not to penalize legitimate engagement events where the user interacts, as this experience can be frustrating to the user, also reduce revenue. With FRL-BSC using Blockchain with federated learning, improving the accuracy of anomaly detection while using distributed intelligence and immutable verification, thus scoring a significant reduction in the FPR by 3.8% made evaluated using Eq. 16. Monitoring and tuning the False Positive Rate can be a valuable tool to enhance security models systematically. To achieve excellent detection of fraud without, at the same time, inhibiting valid ad delivery. Overall, managing the False Positive Rate in the complexity and scale of IoT advertising networks helps to increase the system's dependability, increase user trust, and enhance operational efficiency, as shown in Fig. 8. For Device_001, MCA records the highest FPR at 9.8%, while FRL-BSC achieves the lowest at 5.0%. On Device_002, MCA peaks at 10.2%, IPFS at 9.0%, DNNs-BCT lowers to 5.0%, and FRL-BSC remains efficient at 6.0%. For Device_003, MCA drops to 7.0%,

DNNs-BCT rises to 6.8%, IPFS records 7.7%, and FRL-BSC minimizes further to 3.9%. Finally, on Device_004, MCA stabilizes at 8.0%, DNNs-BCT rises to 9.0%, IPFS falls to 6.0%, while FRL-BSC secures the best rate at just 3.7%.

The learning process is referred to as the local model update from one or more reinforcement learning agents learning from data and outputting incremental training outputs from their learning on IoT devices, using data from their private interactions. The architecture of FRL-BSC uses decentralized training to allow for adaptive updating of ads based on variations in device type, context, and also discretionary time factors by 92.8% made evaluated using Eq. 17. Integrating the update process on the Blockchain will help track changes, in an accountable, transparent, and immutable way, producing secure contextually aware updates of model changes that can reduce communication costs, reduce overhead during inference, and provide a scalable solution that can support personalized real-time advertising to target local user behaviour without data all over the place in Fig. 9. The graph shows model accuracy trends across epochs for MCA, DNNs-BCT, IPFS, and FRL-BSC. At 20 epochs, FRL-BSC leads with 80%, followed by IPFS at 68%, MCA at 60%, and DNNs-BCT lowest at 55%. By 40 epochs, IPFS improves to 75% and MCA to 67%, while FRL-BSC stays steady at 78% and DNNs-BCT drops to 50%. At 60 epochs, FRL-BSC climbs to 85%, IPFS peaks at 78%, MCA dips to 55%, and DNNs-BCT rises to 56%. Finally, at 80 epochs, FRL-BSC achieves the highest accuracy of 93%, MCA recovers to 70%, DNNs-BCT improves to 62%, while IPFS falls back to 60%. Overall, FRL-BSC consistently outperforms all models across training.

In the FRL-BSC framework, the blockchain transaction hash ensures the security of ad-related activities in programming. It consists of a transaction hash to uniquely identify interactions and ad records, a block number to determine the record's location, a

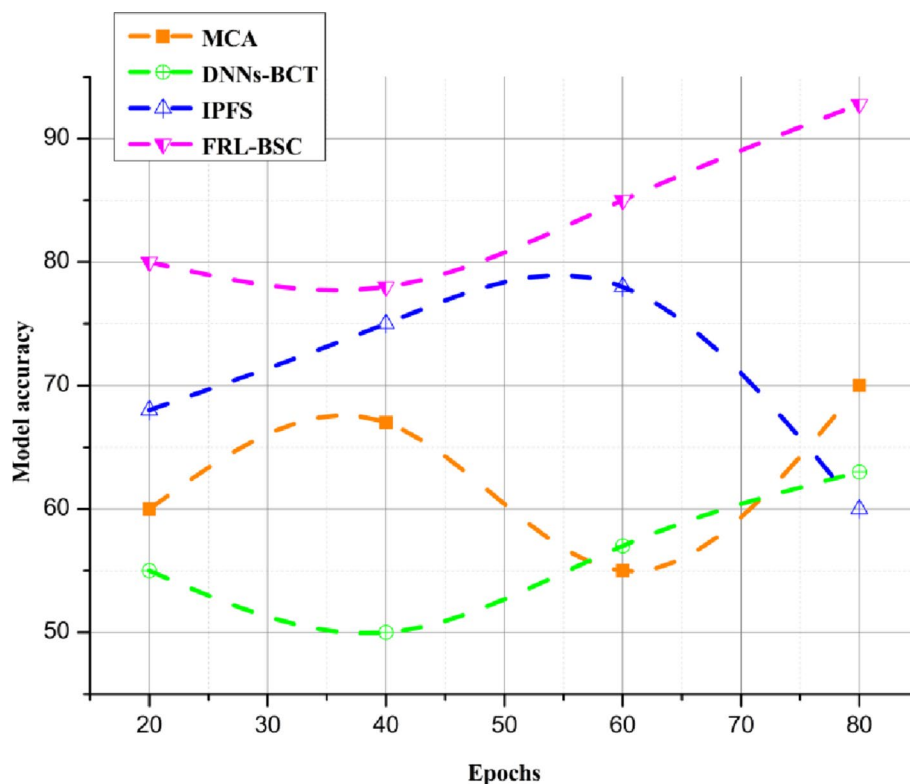


Fig. 9 The analysis of local model update

Timestamp to synchronize with triggered interaction events, and a smart contract ID to associate the interaction with execution logic made evaluated using Eq. 18. Together, this information enables data integrity, transparency, verification of transactions, and trust in decentralized IoT advertising ecosystems. The high-level structure of FRL-BSC principles and properties enables reasonable avenues of best-case scenario investment in Table 4.

Network Latency refers to the time delay that occurs during communication between IoT devices, edge nodes, and blockchain technology. Low latency is vital to the real-time delivery of ads, rapid modelling, model aggregation, and lastly, timely confirmations of transactions on the Blockchain. A seamless user experience relies on a well-designed network to facilitate user experiences, speed up personalization, and ensure consistent, synchronized learning across distributed learning processes by 38% made evaluated using Eq. 19. Lastly, latency measurement enables engineers to develop protocols and allocate resources, ultimately producing scalable and manageable IoT solutions that meet the requirements of time-sensitive applications, such as connected vehicles or smart retail environments, as illustrated in Fig. 10. At Interval 5, MCA again peaks (~82 ms), while DNNs-BCT shows ~68 ms, IPFS ~55 ms, and FRL-BSC just ~45 ms. At Interval 7, MCA maintains high latency (~83 ms), DNNs-BCT reduces to ~70 ms, IPFS to ~65 ms, and FRL-BSC drops further to ~42 ms. Finally, at Interval 9, MCA still lags with ~80 ms, DNNs-BCT measures ~72 ms, IPFS ~65 ms, and FRL-BSC leads with the lowest latency of ~38 ms. Overall, FRL-BSC consistently achieves the lowest latency, while MCA suffers from the highest delays.

Revenue Share and Payment Record signify the transparent allocation and documentation of any advertising revenue among key stakeholders (i.e., advertisers, publishers, and IoT device owners). To maintain trust and fairness throughout the ecosystem, stakeholders must guarantee that payment records are always accurate, immutable, and transparent. FRL-BSC employs blockchain technology to set up smart contracts that distribute advertisement revenues on the consensus mechanism of predetermined agreements, achieving a 94.9% success rate made evaluated using Eq. 20. It is critical to ensure a precise payment method for revenue sharing whenever possible, as this creates incentives for stakeholders to participate in the ecosystem, allows for flexibility in sustainable growth, and engenders further trust in a proven, secure, and decentralized IoT Ford advertising ecosystem in Fig. 11. Table 5 shows Specification of Experimental set up. Table 6 presents comparison of results.

Table 4 Blockchain transaction hash

Transaction Hash	Unique cryptographic identifier for a blockchain transaction	String	0×8fa4b1d93e7a2f4c9c8a...	Ensures immutability and enables quick verification of stored ad interaction or model updates
Block Number	A blockchain block is where the transaction is recorded	Integer	1,745,632	Supports chronological ordering and efficient retrieval of transaction data
Timestamp	The time when the transaction was added to the Blockchain	Data Time	2025-08-11 14:25:47	Matches with interaction events for traceability in ad campaigns
Smart Contract ID	Identifier of the smart contract executing the transaction	String	SC_FRL_BSC_2025_09	Links transaction to contract logic for revenue sharing, model update validation, or access control

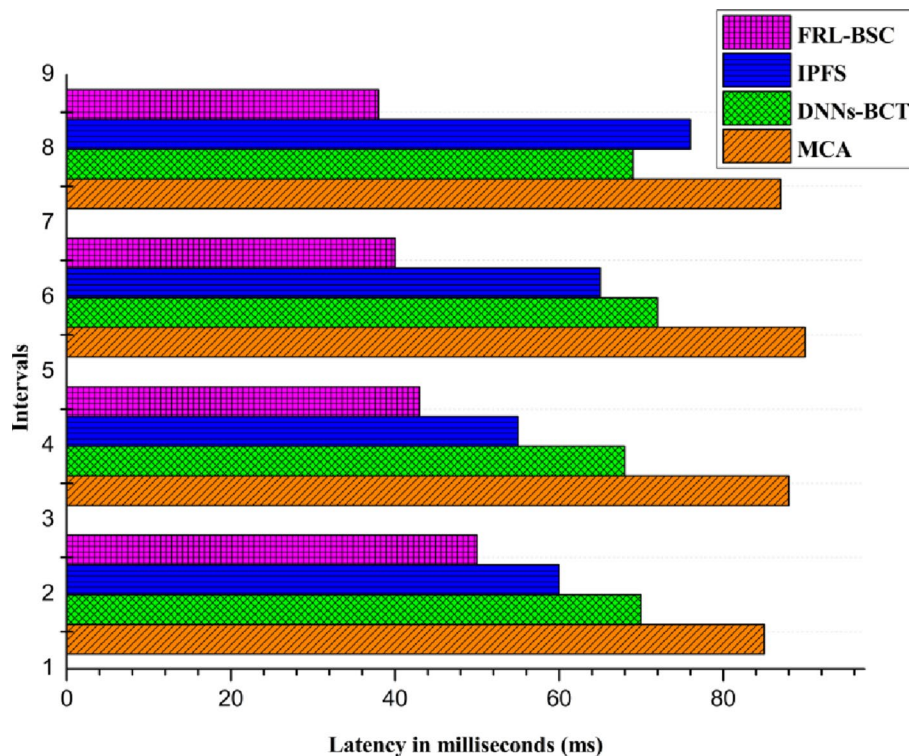


Fig. 10 The analysis of Network Latency

In summary, the assessment shows that FRL-BSC is better than MCA [13], DNNs-BCT [19], and IPFS in every way, including better security, reduced latency, better accuracy, and clear payment records. The framework combines federated learning with Blockchain to ensure that IoT-based digital advertising operates safely, quickly, and reliably. This makes it a strong option for future smart and decentralized marketing ecosystems.

6 Conclusion

The FRL-BSC framework offers a new perspective on IoT-based digital advertising by combining federated reinforcement learning, blockchain smart contracts, and edge computing. The integration of these three domain technologies will help to address issues such as centralized control, privacy violations, ad fraud, and a lack of transparency. Through decentralized model training and immutable recording of transactions, FRL-BSC promotes secure and privacy-preserving personalized ads in real-time. It achieves this by recording revenue in a verifiable manner and conducting decentralized and verifiable sharing of revenue obtained. The experimental results indicate that the framework was practical in performance improvements across several areas: 93.1% security mechanism performance improvement; 8% ad campaign identification performance improvement; 3.8% false positive rate associated with correctly identified ad campaigns; 92.8% local model accuracy; a 38% improvement in network latency; and 94.9% fairness of revenue sharing. These outcomes provide evidence-based confidence that FRL-BSC is a reputable, scalable, and well-optimized framework for supporting the next-generation IoT advertising ecosystem, while building user trust and maximizing operational and commercial value.

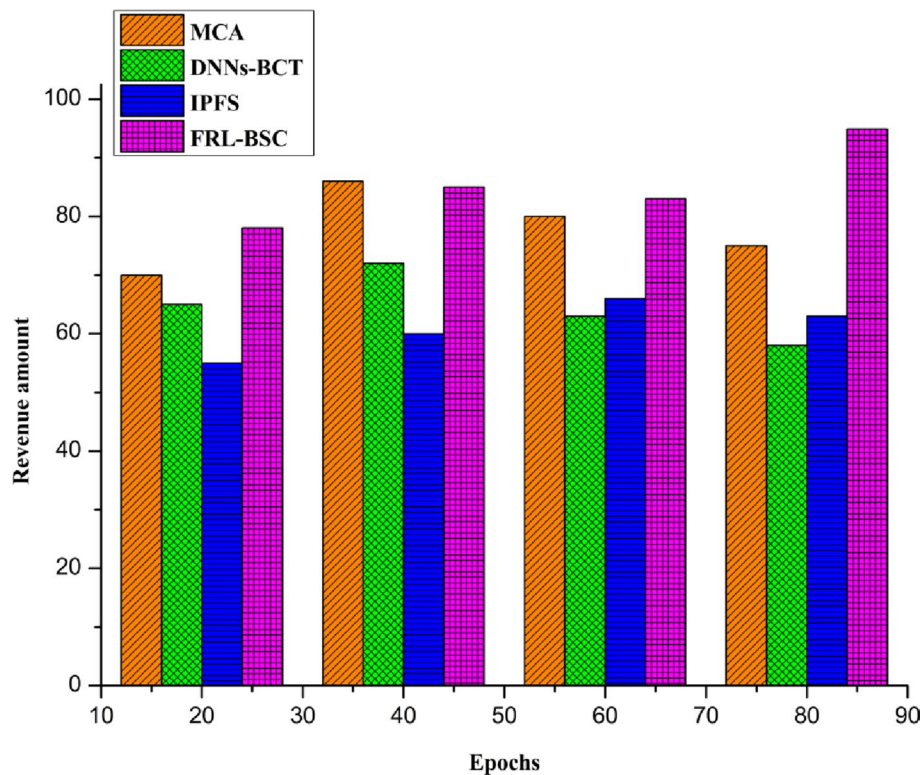


Fig. 11 The analysis of revenue share

Table 5 Specification of experimental set up

Parameter	Description/value
Dataset	e.g., Urban Vehicular Traffic Dataset, 10,000 samples
Number of vehicles/nodes	e.g., 50 simulated vehicles
Hardware configuration	CPU: Intel i7-12700, GPU: NVIDIA RTX 3060, RAM: 32GB
Software environment	Python 3.10, TensorFlow 2.12, NS-3 for simulation
Communication protocol	DSRC/5G/Wi-Fi
Training duration	e.g., 12 h
Testing duration	e.g., 2 h
Evaluation metrics	Accuracy, efficiency, latency, FPR
Baseline methods for comparison	Method A, Method B, Method C

Table 6 Comparison of results

Framework	Accuracy (%)	Latency (ms)	FPR (%)	Energy consumption (J)	Revenue fairness (%)
MCA	87.2	120	8.5	450	78
DNNs-BCT	89.5	105	7.8	470	80
IPFS	85.3	130	9.2	430	76
FRL-BSC	93.8	98	5.1	410	88

Planned future work will entail examining FRL-BSC specifically for cross-domain IoT advertising, developing more advanced privacy-preserving mechanisms, such as homomorphic encryption of model parameters, and extending the framework with adaptive rewards that account for dynamic user behaviour. The zero-knowledge proofs have the potential to augment verification, certainty, and trust, warranting further investigation. They also offer the opportunity to optimize energy consumption in edge nodes, thereby

enhancing the scalability, trust, and sustainability of more complex IoT advertising ecosystems.

Author contributions

Yegen Ouyang: Data Collection and analysis, Writing – review & editing En-wei Cao: Methodology, Investigation, Writing – original draft Bingfeng Liu: Project administration, Writing – review & editing.

Funding

The authors did not receive any financial support.

Data availability

Data are obtained from https://www.kaggle.com/datasets/demodatauk/digital-marketing-eventlevel-sample?utm_source=ce. Dataset name: Digital Marketing Ads & Events—sample.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 13 August 2025 / Accepted: 18 November 2025

Published online: 28 November 2025

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